enjoy the benefits of multi-touch technology built in traditional touchscreen designs in smartphones, tablets and other touch-enabled human-machine interfaces (HMIs). However, the multi-touch application in smartphones and tablets has replaced a keyboard and a mouse to access different screens.

Multi-user multi-touch is a highly intuitive technology that enables a trackpad or touchscreen to simultaneously sense the presence of more than two concurrent points of contact with the surface. This has a profound impact on end-users, panel operators, engineers, managers, purchasers, executives and other decision-makers. With this interaction technique, users can control graphical applications using several fingers, and one or more users can interact with computer applications through various gestures created by fingers on a surface without affecting other users.

A multifaceted experience can be created by simultaneously opening different popups, which encourages greater engagement, sales or training. This versatility enriches the technology for more immediate interaction between the computer and users.

Multi-touch devices are stronger than dual-touch devices but are also expensive because the technology has to support several concurrent contact points, and it also has to sense each touch. In today's market, implementing a multi-touch display for multiusers will enhance customer experiences and, thus, enable entrepreneurs to make strategic business decisions.

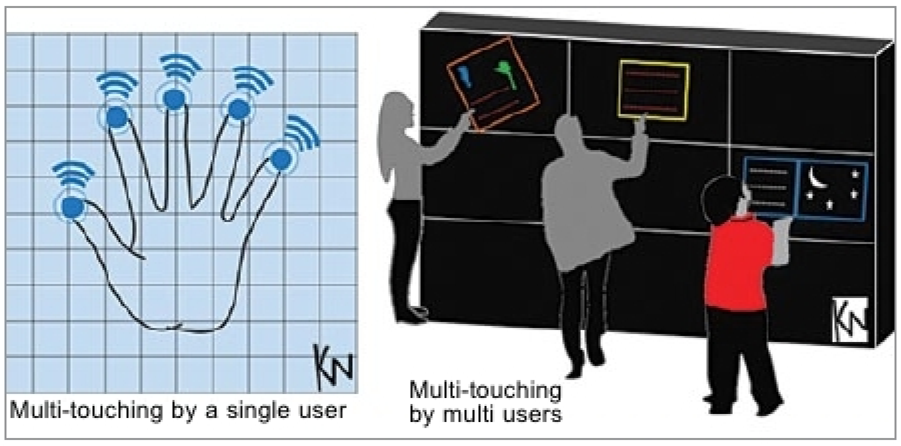

Fig. 3: Single user versus multiple users using the technology

Fig. 4: Multi-touch interactive wall (Credit: eyefactive interactive systems)

Multi-user multi-touch technology is used in multi-touch interactive walls, also called walls of wonders used in public spaces or collaboration environments, like museums, schools, hospitals, hotels, spas, conference rooms of offices, etc that show a real-time 3D atmosphere where users can access over 2000 files such as PDFs, videos and images. These files can be sent via email, and shared and compared by other users. Hence, this technology has proven its significance in improving customer relationships.

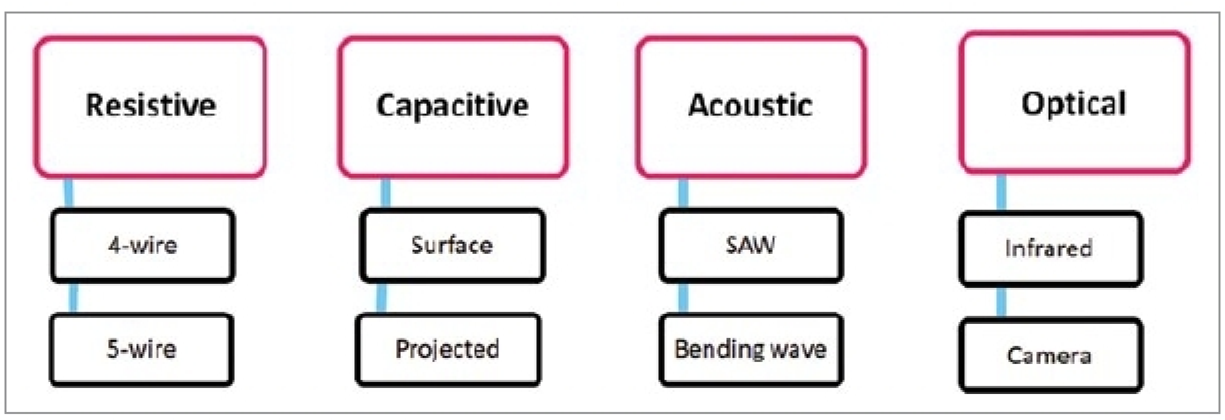

Fig. 5: Classification of multi-touch technology

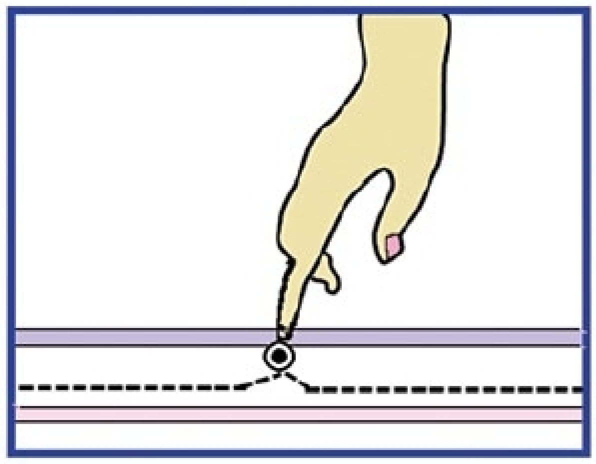

Fig. 6: Projected capacitance

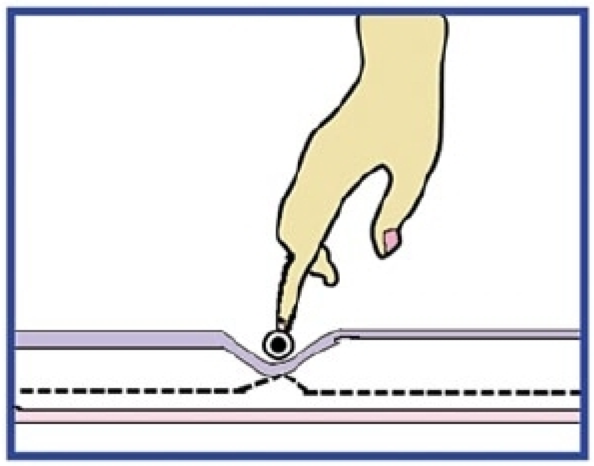

Fig. 7: 3D touch

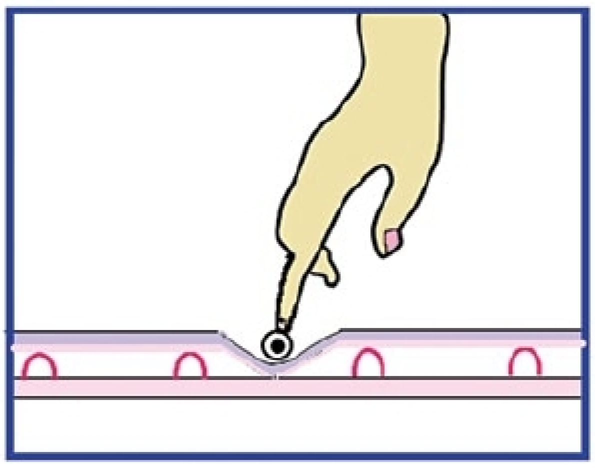

Fig. 8: Resistive touch

The electronics and semiconductor industry pays special focus to the multi-user multi-touch screen market analysis, which is expected to rise by 2027. The classification of multi-touch technology is as follows:

Based on the built-in technology and application, multi-user multi-touchscreen is segmented in the global market.

Based on technology, it is segmented into capacitive, resistive, acoustic, and optical and IR. Each technology is diverse in terms of approach, from problem recognition and interpretation to simultaneous multiple touches.